

CIP Project – Feasibility Study

Analysis and Prediction of Flight Departure Delays at Zurich Airport (ZRH)

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1. Introduction

This project aims to analyze and predict commercial flight departure delays at Zurich Airport (ZRH) using structured aviation data obtained through API requests in Python. The objective is to identify operational factors associated with delays and assess whether departure delay in minutes can be predicted using information available before or at scheduled departure time.

2. Research Questions

- How accurately can departure delay be predicted using operational flight information at Zurich Airport?** A regression model will predict departure delay in minutes and will be evaluated using MAE, RMSE, and R^2 .
- Does route-level historical delay improve prediction accuracy?** A baseline model using flight-specific variables will be compared with an extended model that also includes average historical delay for each route.
- Which operational variables significantly influence departure delay?** Model coefficients and feature importance measures will be interpreted to identify the main drivers of delay.

3. Data Source

Flight data will be collected using the **Apify** platform through structured API requests implemented in Python. The returned JSON data will be transformed into pandas DataFrames. Relevant variables include flight number, airline, origin, destination, scheduled departure time, actual departure time, flight status, and delay duration. If delay is not directly available, it will be computed from the timestamps.

4. Time Range Selection

To keep the project practical and manageable, the analysis will focus on flights departing from Zurich Airport during the period shown in Table 1.

Table 1: Selected observation period

Start Date	End Date	Duration
01 October 2025	31 December 2025	3 months

This time window is sufficiently large to provide a meaningful number of observations while remaining feasible under possible API rate limits. It also captures seasonal variation, including the winter holiday travel period, which may influence delay patterns.

5. Derived Features

Additional explanatory variables will be derived from the scheduled departure timestamp and route information. These variables improve the practical relevance of the analysis and help capture recurring temporal patterns.

Table 2: Planned derived variables

Feature	Description
Departure Hour	Hour extracted from the scheduled departure timestamp to capture peak congestion periods.
Day of Week	Weekday of departure to reflect differences between business and leisure travel patterns.
Month	Calendar month to account for seasonal effects.
Weekend Indicator	Binary variable indicating whether the flight departs on a weekend.
Route-Level Historical Delay	Average historical delay for the same origin–destination route.

6. Methodology

The project will proceed in four steps:

1. **Data acquisition:** retrieve flight data via API requests.
2. **Cleaning and transformation:** inspect missing values, data types, expected ranges, and outliers; derive additional variables.
3. **Analysis and visualization:** explore delay patterns across airlines, routes, and time periods using clear figures and summary tables.
4. **Modeling:** build and compare regression models for delay prediction.

7. Risk Analysis

Table 3: Main risks and mitigation strategies

Risk	Description	Mitigation
API rate limits	Free access may restrict the number of requests.	Restrict the observation period and cache responses locally.
Missing records	Some flights may lack actual departure timestamps or delay values.	Filter incomplete observations and document missing-value handling.
Outliers	Extreme or negative delay values may distort results.	Identify implausible values, separate cancellations, and treat outliers transparently.

8. Conclusion

Overall, with this proposal, we can explore a practical application of Python to a real-world transportation problem. It includes all required components of CIP: API-based data acquisition, structured cleaning and transformation, visualization, and predictive modeling.